روش‌های پیشرفته نمونه برداري در جنگل

مقدمات

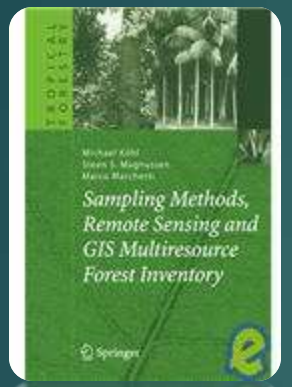
هرمز سهرابي، دانشکده منابع طبيعي دانشگاه تربيت مدرس

مدل مفهومي نمونه برداري

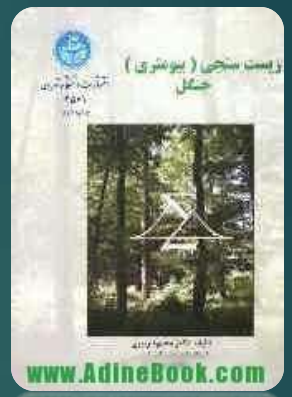
- Protocol

- طراحي نمونه برداري — Sampling design
  - روش نمونه برداري — Sampling Method
    - نمونه برداري نقطه‌اي Point Sampling
    - نمونه برداري فاصله‌اي Distance Sampling
    - نمونه برداري خطي Line Sampling
    - نمونه برداري قطعه‌اي Plot Sampling
  - الگوي نمونه برداري — Sampling patterns
    - تصادفي ساده Simple Random S.
    - بلوك تصادفي Block Random S.
    - سيستماتيک تصادفی Systematic S.
  - اجتماع نمونه — Sampling clusering
    - نمونه واحد Single s.
    - نمونه خوشه‌اي Cluster S.
  - استراتژي نمونه برداري — Sampling strategies
    - مونه بندي
      - پيشا مونه بندي(Prestratification)
      - پسا مونه بندي (Poststratification)
    - نمونه برداري چند گانه — Multiphase S.
      - Two phase sampling with regression estimator
      - Two phase sampling with ratio estimator
      - Two phase sampling for stratification
      - .....
    - نمونه برداري چند گامي — Multistage S.
      - Two stage sampling
      - Three stage sampling
      - .....

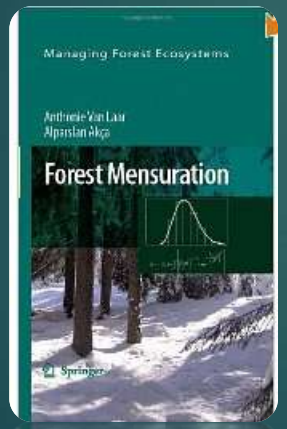
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Sampling Methods, Remote Sensing And GIS Multiresource Forest Inventory
***Kohl et al*. 2006**
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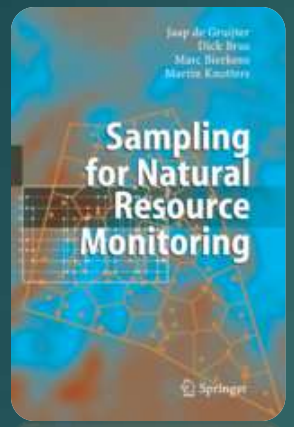


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زبيري، 1386
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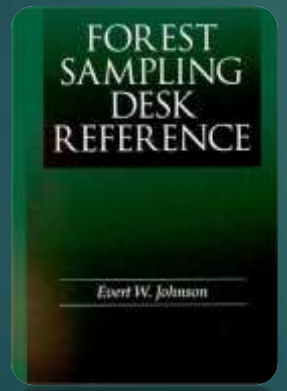
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***Laar and Akca*, 2007**
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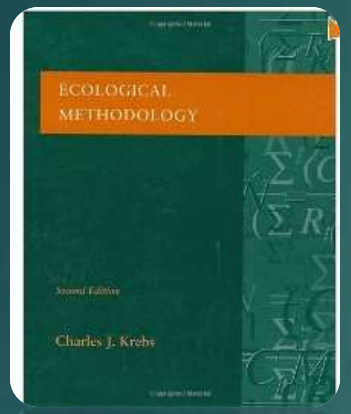
Sampling for Natural Resource Monitoring
Gruijter **et al. 2006**
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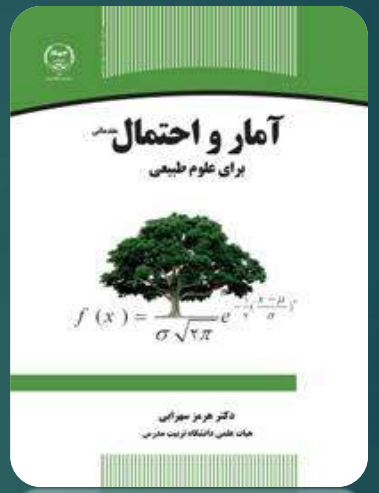


Forest Sampling Desk Reference
Johnson**, 2000**
CRC Press.

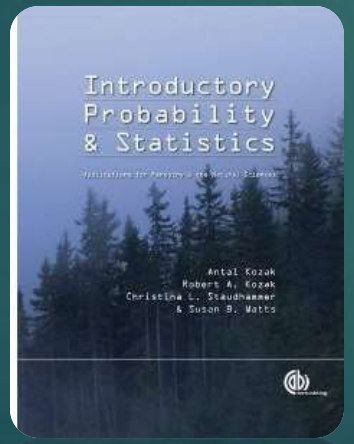
منابع



Ecological Methodology
***Krebs*, 1998**
Benjamin Cummings.



آمار و احتمال مقدماتي براي علوم طبيعي
***سهرابي*، 1391**
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Introductory probability and statistics
Kozak, et al., 2008
CABI Press.

TERMINOLOGY

- جمعيت آماري (Statistical Population)
- واحد نمونه برداري (Sampling Unit)
- نمونه (Sample)
- پارامتر (Parameter) و آماره (Statistics)
- اندازه گيري (Measurement)

Reasons why sampling may be preferred over complete enumeration

1. **It may be impossible to obtain a complete enumeration.** For example, if the evaluation process is destructive

2. **A complete enumeration may be too expensive.** Insofar as budgets are concerned, inventories are cost items and usually the funds available for such activities are limited. Only rarely are such funds sufficient for a complete enumeration.
3. **A complete enumeration may require more time than is available**

4. **In some cases sampling may actually yield a more reliable final value than would a complete enumeration.** Since the time requirement for sampling is less than that for a complete inventory, more time can be taken to obtain each individual evaluation, and this usually results in better or stronger individual values. Furthermore, the cost of sampling is less than that of a complete evaluation and the saving can be used to improve the instrumentation.

پارامتر و آماره متناظر

برآورد آماري (Statistical Estimation)

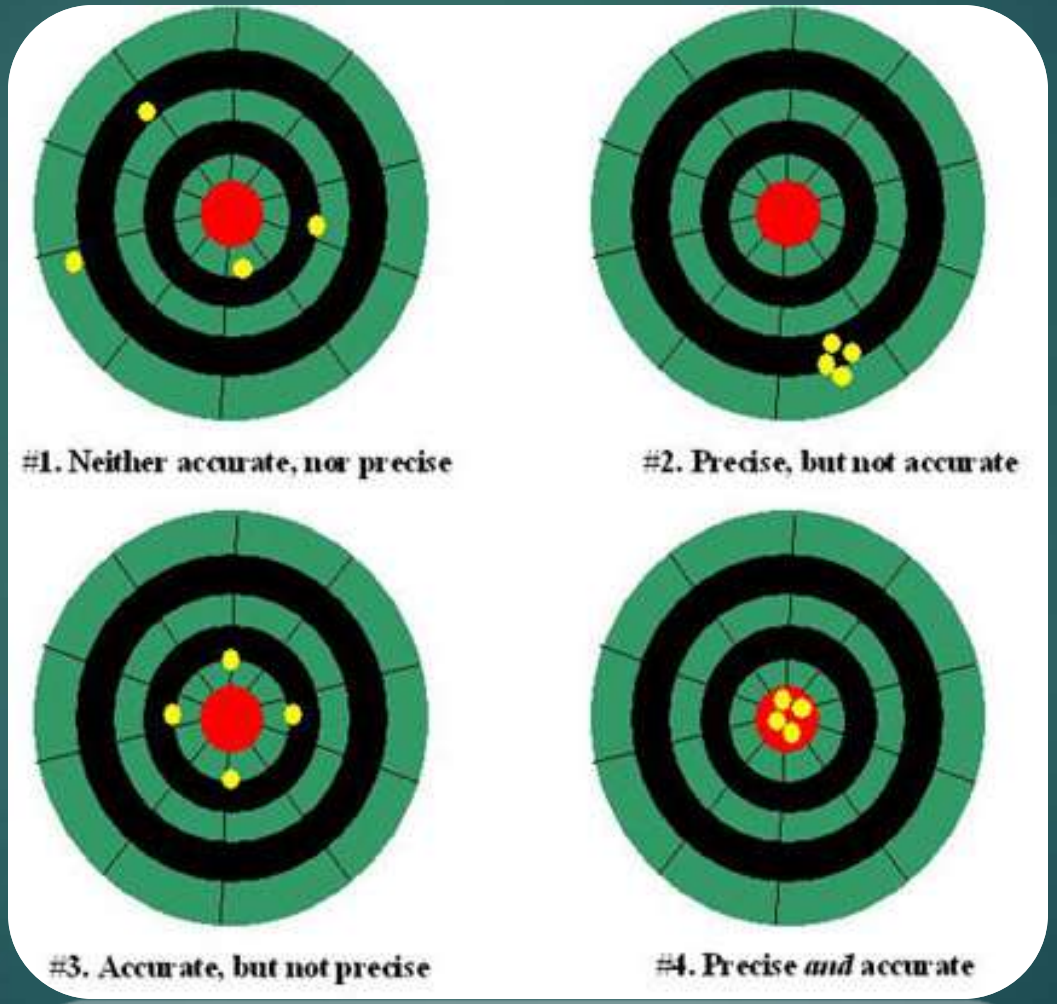
شاخص	میانگین	واریانس	نسبت	ضریب همبستگی
نماد پارامتر (جمعیت)	μ	σ^2	π	ρ
نماد آماره (نمونه)	$\bar{x}$	s^2	p	r

Major error types :

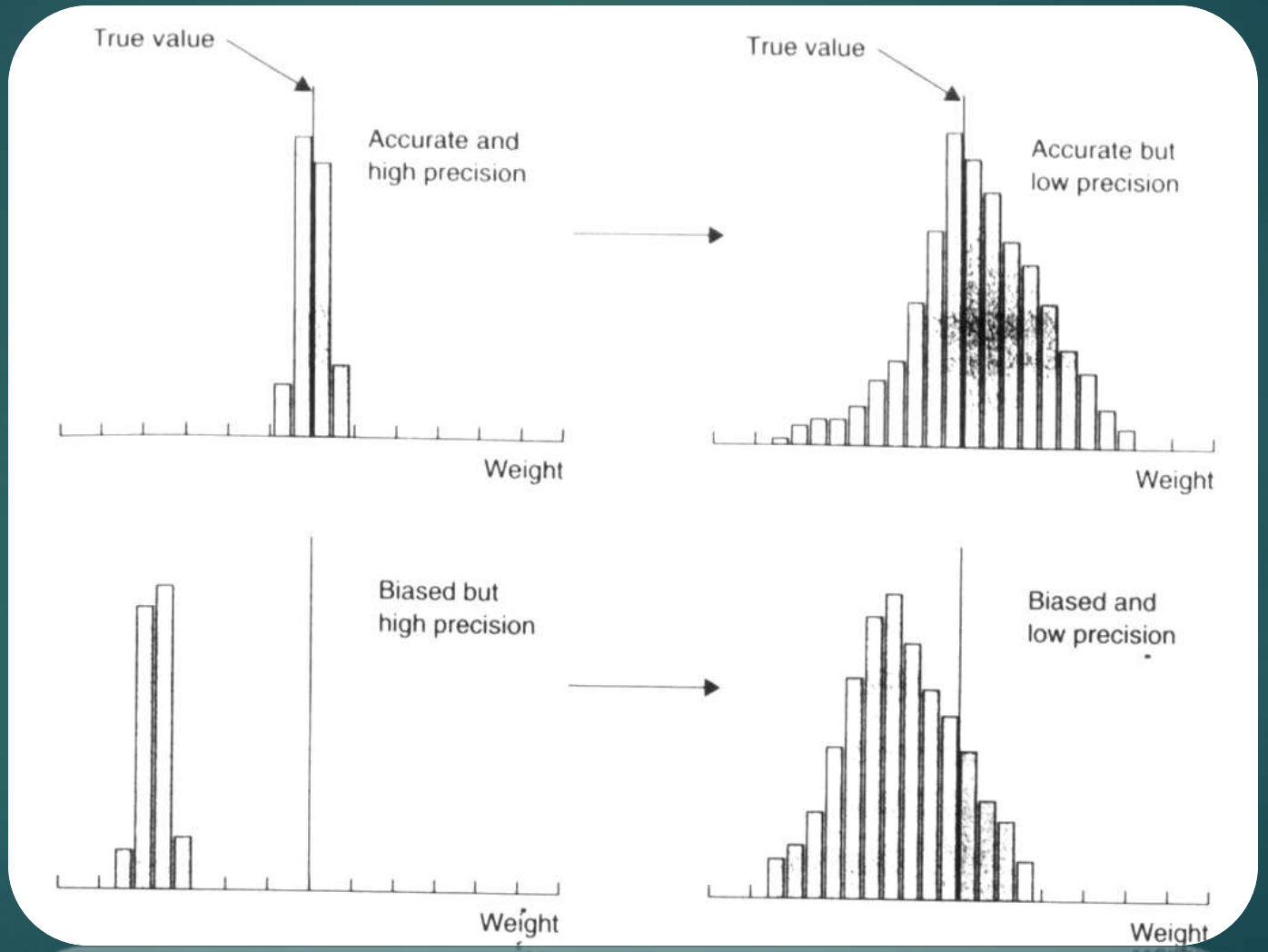
1–*Sampling Error*

2–*Nonsampling Error* :*random and nonrandom errors*

دقت (Precision) و صحت (Accuracy)



دقت (Precision) و صحت (Accuracy)



تعداد ارقام اعشار

- **Sokal and Rohlf (1995) Method**

$$\frac{Range}{30} = Minimal\ desired\ level\ of\ \ measurement$$

$$\frac{Range}{300} = Maximal\ desired\ level\ of\ \ measurement$$

$example:$

$$10 < DBH\ < 100 \rightarrow Range = 90$$

$$\frac{90}{30} = 3$$

$$\frac{90}{300} = 0.3$$

منابع این جلسه

Forest Sampling Desk Reference
***Johnson*, 2000**
CRC Press.

Chapter 12: sampling theory
Chapter 13: Estimation of parameters

Introductory probability and statistics
Kozak, et al., 2008
CABI Press.

Chapter 8: Estimation